Lab 05 — AutoEncoders

MNIST Dataset Compression and Reconstruction

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Objectives

1. Train a **PCA** model to compress MNIST to a **4-dimensional** latent vector.
2. Train an **AutoEncoder (AE)** neural network to compress MNIST to a **4-dimensional** latent vector.
3. Visualize latent vectors using **t-SNE** for both models.
4. Show **20 reconstructions** from latent samples for both models.
5. Compute and compare the **compression ratio**.

Dataset: MNIST

MNIST contains 70,000 grayscale 28×28 digit images (0–9): 60,000 train, 10,000 test. Each image is flattened to **784** dimensions and pixel values normalized to $[0, 1]$.

Method 1 — PCA (Linear Compression)

Theory

PCA projects data onto the k eigenvectors of the covariance matrix with the largest eigenvalues:

$$\mathbf{z} = \mathbf{W}^\top(\mathbf{x} - \boldsymbol{\mu}), \qquad \hat{\mathbf{x}} = \mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \qquad \mathbf{W} \in \mathbb{R}^{784\times 4}$$

It is a **linear** method — the latent space is a linear subspace of $\mathbb{R}^{784}$.

Results

Component	PC1	PC2	PC3	PC4
Explained Variance	9.70%	7.10%	6.17%	5.39%
Total explained variance with 4 PCs: 28.36%				

With only 4 components, PCA retains **28.36%** of the total data variance — meaning **71.64% of information is discarded**, which directly causes the blurry reconstructions observed.

Method 2 — AutoEncoder (Non-linear Compression)

Architecture

$$784 \xrightarrow{\text{BN+ReLU}} 256 \xrightarrow{\text{BN+ReLU}} 64 \rightarrow \underbrace{4}_{\text{bottleneck}} \xrightarrow{\text{BN+ReLU}} 64 \xrightarrow{\text{BN+ReLU}} 256 \xrightarrow{\sigma} 784$$

Total parameters: 437,396. Loss: MSE. Optimizer: Adam ($lr = 10^{-3}$, $\lambda = 10^{-5}$). Epochs: 30. Batch: 256.

Training Results

Epoch	Train Loss	Test Loss
1	0.05039	0.03658
5	0.03119	0.03073
10	0.02986	0.02937
15	0.02876	0.02850
20	0.02854	0.02827
25	0.02793	0.02789
30	0.02776	**0.02779**

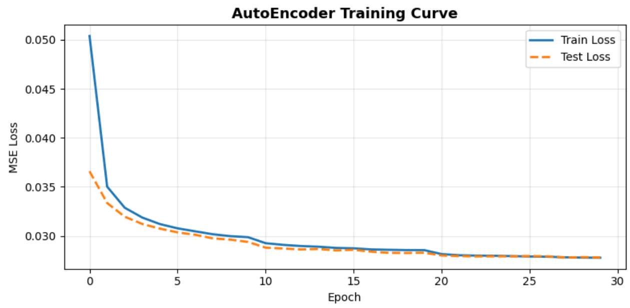

Figure 1: AE training & validation MSE loss over 30 epochs.

The train and test loss curves converge closely, confirming the model generalizes well with **no overfitting**.

t-SNE Visualization of Latent Spaces

t-SNE reduces 4D latent vectors to 2D while preserving local neighbourhood structure, run on **5,000 subsampled test points**.

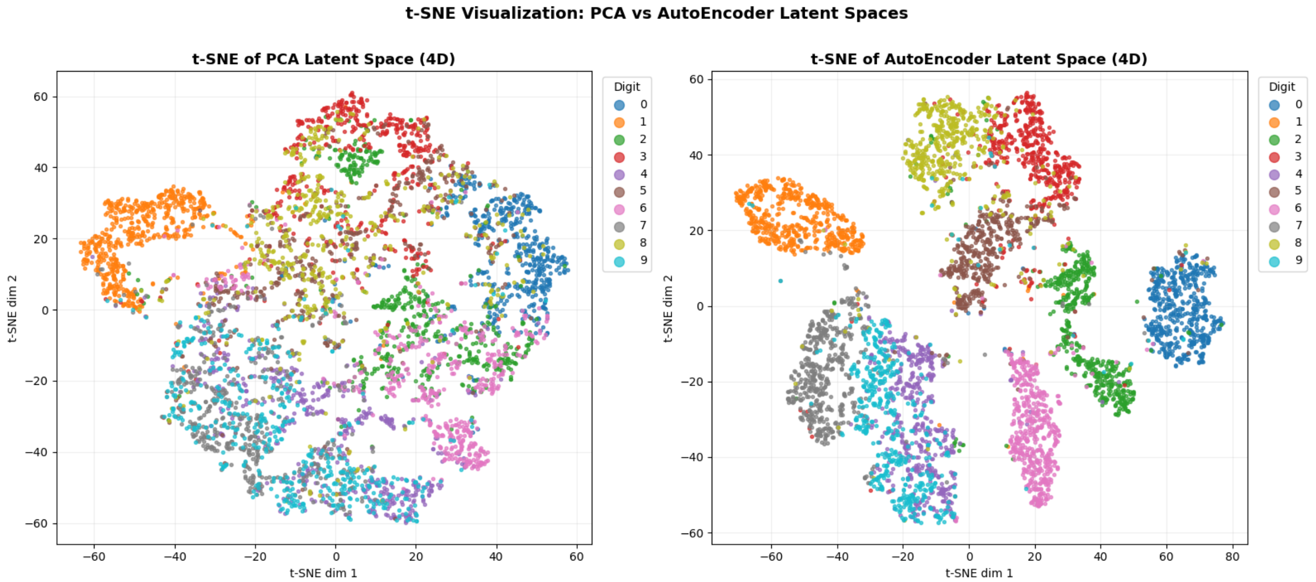

Figure 2: t-SNE of PCA latent space (left) vs AE latent space (right), coloured by digit class.

Observation: The PCA latent space shows significant cluster *overlap* — particularly among visually similar digits (3/5, 4/9, 7/1). In contrast, the AE latent space produces **tighter, more separated clusters** for all 10 digit classes, confirming that the non-linear AE extracts far more discriminative features in the same 4 dimensions.

20-Sample Reconstruction

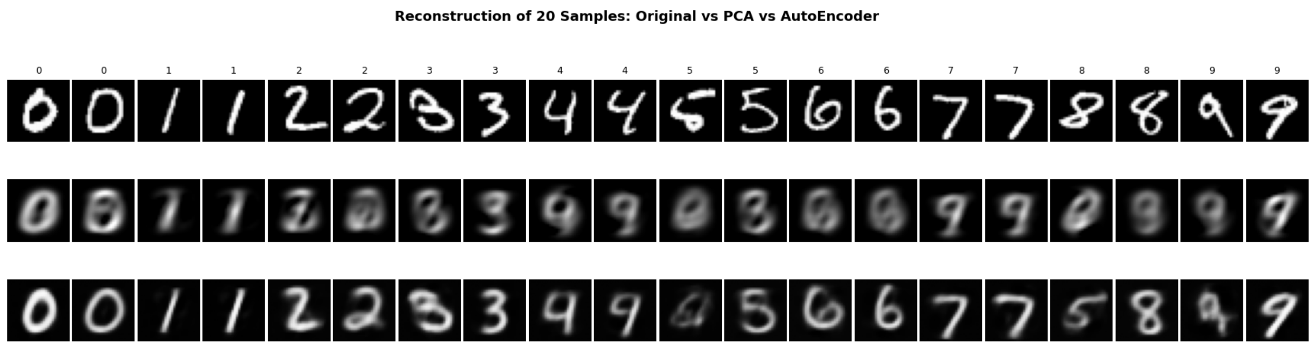

Figure 3: Row 1: Original. Row 2: PCA reconstruction. Row 3: AE reconstruction. Two samples per digit (0–9).

Observation: PCA reconstructions are visibly blurry and lose fine stroke details (e.g. the loops in 8, serifs in 1). The AE reconstructions preserve significantly more structural clarity. Some confusion remains at extreme compression (e.g., digit 4 reconstructed as 9), which is expected at 196× compression.

Pairwise Latent Dimension Projections

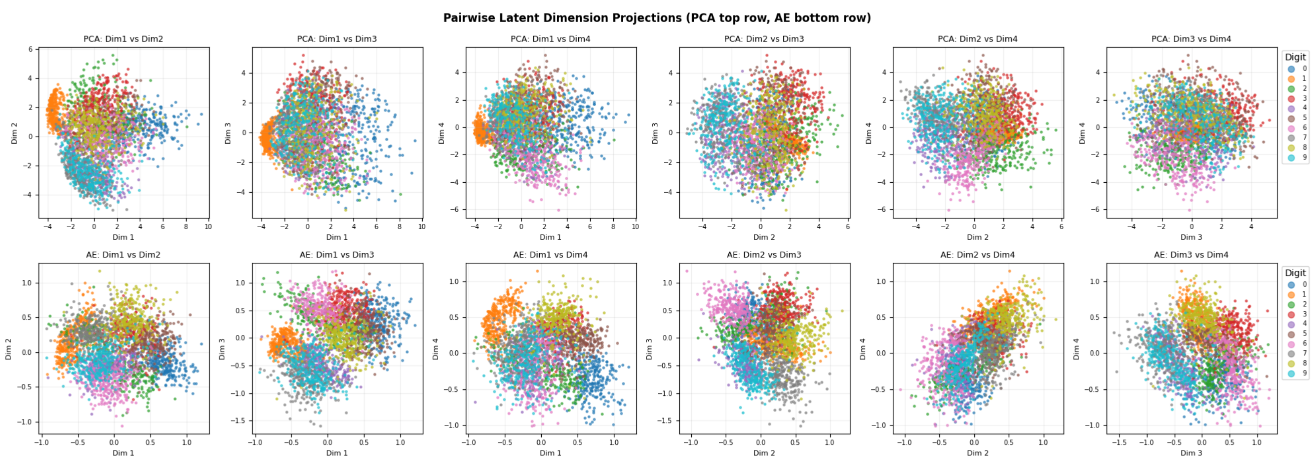

Figure 4: All 6 pairwise projections of the 4D latent space. PCA (top row) vs AE (bottom row).

PCA dimensions are uncorrelated by construction (orthogonal eigenvectors), so clusters span a wider range. AE dimensions are more compressed in scale ($\approx [-1.5, 1.5]$) but encode more class-discriminative structure.

Compression Ratio & Quantitative Comparison

$$\text{Compression Ratio} = \frac{784}{4} = \mathbf{196}\times \qquad \text{Dimension Reduction} = \left(1 - \frac{4}{784}\right) \times 100 = \mathbf{99.5}\%$$

Model	Type	Latent Dim	Compression	Test MSE	Expl. Var.
PCA	Linear	4	196×	0.04737	28.36%
AutoEncoder	Non-linear FC	4	196×	**0.02779**	—

AE achieves **41.3% lower MSE** than PCA at the same compression ratio.

Criterion	PCA	AutoEncoder
Transformation	Linear (eigenvectors)	Non-linear (ReLU activations)
Training time	$<$ 1 second (closed form)	$\sim$2 min (CPU, 30 epochs)
Parameters	$784 \times 4 = 3{,}136$	437,396
Test MSE	0.04737	**0.02779**
Per-sample MSE	0.05219	**0.03521**
t-SNE clusters	Overlapping	Well-separated
Reconstruction	Blurry	Sharper, more detail
Compression	196×	196×

Conclusions

1. Both PCA and the AutoEncoder achieve identical **196× compression** (784 → 4 dimensions, 99.5% reduction).
2. **PCA** with 4 components captures **28.36%** of the total variance. Reconstructions are blurry as the majority of image information is discarded.
3. The **AutoEncoder** achieves a test MSE of **0.02779** — a **41.3% improvement** over PCA (MSE 0.04737) at the same compression. Non-linear ReLU activations allow it to learn curved manifolds in latent space.
4. **t-SNE** confirms the qualitative advantage: AE latent codes form compact, well-separated digit clusters, while PCA clusters show substantial overlap — especially for visually similar pairs (3/5, 7/1, 4/9).
5. The full pipeline (data loading, PCA, AE training ×30 epochs, t-SNE, all visualizations) completes in **under 5 minutes on CPU** — no GPU required.

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